

Shivankur Ghaie

Capstone Project 2

1. Creation of storage account:

raw

Container

Search

0 <<

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

+ Add Directory

↑ Upload

↻ Refresh

🗑 Delete

📄 Copy

📄 Paste

🔄 Rename

🔑 Acquire lease

🔑 Break lease

📄 Edit columns

raw

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 4 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	__unitystorage	26/8/2025, 2:23:54 PM				...
☐	__schemas	26/8/2025, 2:48:09 PM				...
☐	products	26/8/2025, 10:35:21 AM				...
☐	sales	26/8/2025, 10:35:38 AM				...

File shares

Queues

Tables

Security + networking

Data management

Settings

Monitoring

Monitoring (classic)

Blob anonymous access

Disabled

Blob soft delete

Enabled (7 days)

Container soft delete

Enabled (7 days)

Versioning

Disabled

Change feed

Disabled

NFS v3

Disabled

SFTP

Disabled

Storage tasks assignments

None

File service

Storage account key encryption

Enabled

Minimum TLS version

Version 1.2

Infrastructure encryption

Disabled

Networking

Public network access

Enabled

Public network access scope

Enable from all networks

Private endpoint connections

0

Network routing

Microsoft network routing

Endpoint type

Standard

2. Creation of access connector:

AccessConnectorCreate-20250826101243

Overview

Deployment

Search

×

<<

🗑 Delete

🔄 Cancel

🔄 Redeploy

⬇ Download

↻ Refresh

Overview

Inputs

Outputs

Template

🟢 Your deployment is complete

Deployment name : AccessConnectorCreate-20250826101243

Subscription : LabsKraft

Resource group : LabsKraft2027

Start time : 26/8/2025, 10:14:12 AM

Correlation ID : fbf037fd-a483-407c-8ec1-76ffc34168ee

> Deployment details

> Next steps

Go to resource

3. Giving access connector permission for storage:

Add role assignment ...

[Role](#) [Members](#) [Conditions](#) [Review + assign](#)

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)

Job function roles [Privileged administrator roles](#)

Grant privileged administrator access, such as the ability to assign roles to other users.

⚠ Can a job function role with less access be used instead?

🔍 Search by role name, description, permission, or ID

Type : All

Category : All

Name ↑↓	Description ↑↓	Type ↑↓	Category ↑↓	Details
Contributor	Grants full access to manage all resources, but does not allow you to assign roles in Azure R...	BuiltInRole	General	View
Azure File Sync Administrator	Provides full access to manage all Azure File Sync (Storage Sync Service) resources.	BuiltInRole	None	View
Azure Resilience Management Drills...	Administrator Role of Azure Resilience Management Drills Service	BuiltInRole	None	View

Showing 1 - 3 of 3 results.

Name * (🔍)

shiv connector

Add role assignment ...

[Role](#) [Members](#) [Conditions](#) [Review + assign](#)

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)

Job function roles [Privileged administrator roles](#)

Grant access to Azure resources based on job function, such as the ability to create virtual machines.

🔍 storage blob con

Type : All

Category : All

Name ↑↓	Description ↑↓	Type ↑↓	Category ↑↓	Details
Storage Blob Data Contributor	Allows for read, write and delete access to Azure Storage blob containers and data	BuiltInRole	Storage	View
Storage Blob Data Owner	Allows for full access to Azure Storage blob containers and data, including assigning POSIX a...	BuiltInRole	Storage	View
Storage Blob Data Reader	Allows for read access to Azure Storage blob containers and data	BuiltInRole	Storage	View

Showing 1 - 3 of 3 results.

Add role assignment ...

[Role](#) [Members](#) [Conditions](#) [Review + assign](#)

🔍 Showing a filtered list of roles because your permissions include a condition. [Learn more](#)
[View my access](#)

Selected role Contributor

Assign access to ☐ User, group, or service principal
☒ Managed identity

Members + Select members

Name	Object ID	Type
No members selected		

Description Optional

Select managed identities

⚠ Some results might be hidden due to your ABAC condition.

Subscription *

LabsKraft

Managed identity

Access Connector for Azure Databricks (12)

🔍 shiv



shivamconnector
/subscriptions/d008c7cc-9795-4292-bf79-74ae52cda63d/resourceGroups/LabsKra...

Selected members:



shiv_connector
/subscriptions/d008c7cc-9795-4292-bf79-74ae52cda63d/resourceGroups/... [Remove](#)

4. Creation of Unity Catalog:

Create a new catalog

A catalog is the first layer of Unity Catalog's three-level namespace and is used to organize your data assets. [Learn more](#)

Catalog name*

cap_2_shiv_catalog

Type*

Standard

Storage location

Select external location

sub/path

Create a new external location

Location in cloud storage where data for managed tables will be stored. If not specified, the location will default to the metastore root location.

Cancel

Create

Catalog Explorer >

cap2_shiv_catalog

Share

Create schema

Overview

Details

Permissions

Workspaces

Description

AI generate

Add

Filter schemas

3 schemas

Name	Owner	Created at
cap2_shiv_schema	tredence19@michaelcloudkrafthotm...	Aug 26, 2025 at 10:44 AM
default	tredence19@michaelcloudkrafthotm...	Aug 26, 2025 at 10:42 AM
information_schema	System user	Aug 26, 2025 at 10:42 AM

About this catalog

Owner

Tredence LabsKraft19

Tags

Add tags

5. Adding User to Unity Catalog

Workspace settings > Identity and access >

Users

Q Filter users

17 total

Add user

Status	Email	Name
✓	asanyam10@gmail.com	
✓	bajjijis870@amcret.com	
✓	hacono4632@aperiol.com	
✓	michael.kloudkraft@hotmail.com	Michael A
✓	rohit@michaelkloudkrafthotmail.onmicrosoft.com	RohitY
✓	shibangdas3@gmail.com	
✓	shivamam26@gmail.com	
✓	shivankurpgw1@gmail.com	
✓	shivankurpgw@gmail.com	
✓	shreyas.griduser@gmail.com	
✓	shreyastiware7705@gmail.com	shreyas tiwari
✓	tredence17@michaelkloudkrafthotmail.onmicrosoft.com	Tredence LabsKraft17
✓	tredence18@michaelkloudkrafthotmail.onmicrosoft.com	Tredence LabsKraft18
✓	tredence19@michaelkloudkrafthotmail.onmicrosoft.com	Tredence LabsKraft19
✓	tredence20@michaelkloudkrafthotmail.onmicrosoft.com	Tredence LabsKraft20
✓	tredence21@michaelkloudkrafthotmail.onmicrosoft.com	Tredence LabsKraft21

6. Granting Access to Users

New

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

Catalog

Starter Warehouse Pro S

Type to search...

My organization

tred_d5

system

cap1_shiv_catalog

cap2_shiv_catalog

cap2_shiv_schema

default

information_schema

shiv

shiv_catalog

shibang

shibangcatalog

shivam_catalog_new

Catalog Explorer >

cap2_shiv_catalog

Share

Create schema

Overview

Details

Permissions

Workspaces

Grant

Revoke

Privileges

Type to filter by principal

Principal	Privilege	Object
shivankurpgw1@gmail.com	APPLY TAG	cap2_shiv_catalog
shivankurpgw1@gmail.com	BROWSE	cap2_shiv_catalog
shivankurpgw1@gmail.com	CREATE FUNCTION	cap2_shiv_catalog
shivankurpgw1@gmail.com	CREATE MATERIALIZED VIEW	cap2_shiv_catalog
shivankurpgw1@gmail.com	CREATE MODEL	cap2_shiv_catalog
shivankurpgw1@gmail.com	CREATE SCHEMA	cap2_shiv_catalog
shivankurpgw1@gmail.com	CREATE TABLE	cap2_shiv_catalog
shivankurpgw1@gmail.com	CREATE VOLUME	cap2_shiv_catalog

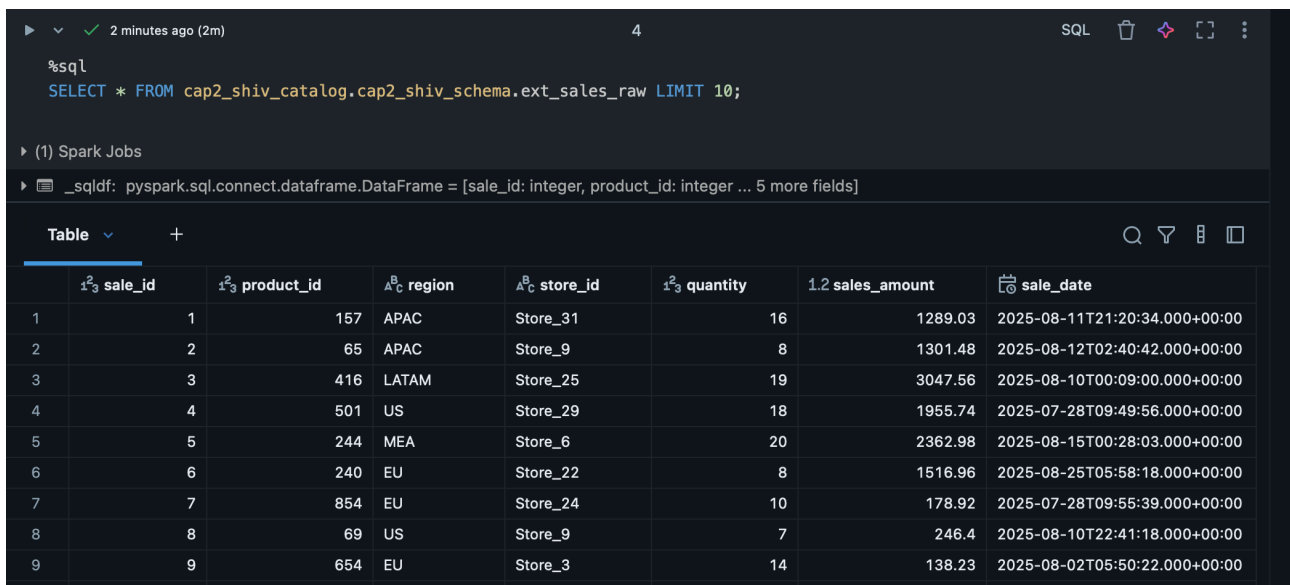
Ingestion

1. Schema Creation:

```
%sql
USE CATALOG cap2_shiv_catalog;

%sql
CREATE SCHEMA IF NOT EXISTS cap2_shiv_schema
COMMENT 'Capstone 2 Schema';

%sql
CREATE TABLE cap2_shiv_catalog.cap2_shiv_schema.ext_sales_raw
USING CSV
OPTIONS (
  path 'abfss://raw@shivghaie.dfs.core.windows.net/sales/',
  header 'true',
  inferSchema 'true'
);
```



The screenshot shows a SQL query execution interface. At the top, there's a status bar with a green checkmark, '2 minutes ago (2m)', and a '4' icon. Below this, the SQL query is displayed: `%sql SELECT * FROM cap2_shiv_catalog.cap2_shiv_schema.ext_sales_raw LIMIT 10;`. Under the query, it says '(1) Spark Jobs'. Below that, a message indicates the query was successful: `_sqldf: pyspark.sql.connect.dataframe.DataFrame = [sale_id: integer, product_id: integer ... 5 more fields]`. The main part of the screenshot is a table with 9 rows and 8 columns. The columns are: `sale_id`, `product_id`, `region`, `store_id`, `quantity`, `sales_amount`, and `sale_date`. The data is as follows:

	<code>sale_id</code>	<code>product_id</code>	<code>region</code>	<code>store_id</code>	<code>quantity</code>	<code>sales_amount</code>	<code>sale_date</code>
1	1	157	APAC	Store_31	16	1289.03	2025-08-11T21:20:34.000+00:00
2	2	65	APAC	Store_9	8	1301.48	2025-08-12T02:40:42.000+00:00
3	3	416	LATAM	Store_25	19	3047.56	2025-08-10T00:09:00.000+00:00
4	4	501	US	Store_29	18	1955.74	2025-07-28T09:49:56.000+00:00
5	5	244	MEA	Store_6	20	2362.98	2025-08-15T00:28:03.000+00:00
6	6	240	EU	Store_22	8	1516.96	2025-08-25T05:58:18.000+00:00
7	7	854	EU	Store_24	10	178.92	2025-07-28T09:55:39.000+00:00
8	8	69	US	Store_9	7	246.4	2025-08-10T22:41:18.000+00:00
9	9	654	EU	Store_3	14	138.23	2025-08-02T05:50:22.000+00:00

```
%sql
CREATE TABLE cap2_shiv_catalog.cap2_shiv_schema.ext_products
USING JSON
OPTIONS (
  path 'abfss://raw@shivghaie.dfs.core.windows.net/products/',
  multiline 'false'
);
```

Medallion Architecture

(i) Bronze Layer

```
import dlt
from pyspark.sql.functions import *
# Ingest raw sales data (multiple daily CSV files in raw/sales/)
@dlt.table(
    name="sales_raw",
    comment="Raw sales data ingested from CSV files",
    table_properties={"quality": "bronze"}
)
def sales_raw():
    return (
        spark.readStream.format("cloudFiles") # Auto Loader for incremental loads
        .option("cloudFiles.format", "csv")
        .option("header", "true")
        .load("abfss://raw@shivghaie.dfs.core.windows.net/sales/")
    )
# Products CSV
@dlt.table(
    name="pro_csv",
    comment="Raw product data from CSV",
    table_properties={"quality": "bronze"}
)
def pro_csv():
    return (
        spark.readStream.format("cloudFiles")
        .option("cloudFiles.format", "csv")
        .option("header", "true")
        .load("abfss://raw@shivghaie.dfs.core.windows.net/products/")
    )
# Products JSON
@dlt.table(
    name="pro_json",
    comment="Raw product data from JSON",
    table_properties={"quality": "bronze"}
)
def pro_json():
    return (
        spark.readStream.format("cloudFiles")
        .option("cloudFiles.format", "json")
        .load("abfss://raw@shivghaie.dfs.core.windows.net/products/")
    )
# Unified Products (CSV + JSON)
@dlt.view(name="products")
def products():
    return dlt.read("pro_csv").unionByName(dlt.read("pro_json"), allowMissingColumns=True)
# Clean Sales
@dlt.table(
```

```

name="sales_clean",
comment="Cleaned sales data with enforced schema",
table_properties={"quality": "silver"}
)

```

(ii) Silver Layer

```

@dlt.expect("valid_sales_amount", "sales_amount >= 0")
@dlt.expect("valid_quantity", "quantity > 0")
def sales_clean():
    return (
        dlt.read("sales_raw")
        .withColumn("sale_id", col("sale_id").cast("string"))
        .withColumn("product_id", col("product_id").cast("string"))
        .withColumn("region", col("region").cast("string"))
        .withColumn("store_id", col("store_id").cast("string"))
        .withColumn("quantity", col("quantity").cast("int"))
        .withColumn("sales_amount", col("sales_amount").cast("double"))
        .withColumn("sale_date", to_date(col("sale_date"), "yyyy-MM-dd"))
        .na.drop()
    )
@dlt.table(
    name="products_clean",
    comment="Cleaned products data with schema evolution support",
    table_properties={"quality": "silver"}
)
def products_clean():
    return (
        dlt.read("products")
        .withColumn("product_id", col("product_id").cast("string"))
        .withColumn("product_name", col("product_name").cast("string"))
        .withColumn("category", col("category").cast("string"))
        .withColumn("price", col("price").cast("double"))
        .withColumn("discount_rate", col("discount_rate").cast("double"))
    )

```

(iii) Gold Layer

```

# Join sales + products and calculate KPIs
@dlt.table(
    name="sales_summary",
    comment="Daily sales summary by product and region",
    table_properties={"quality": "gold"}
)
def sales_summary():
    sales = dlt.read("sales_clean")
    products = dlt.read("products_clean")
    return (
        sales.join(products, "product_id", "left")
        .groupBy("region", "sale_date", "product_id", "product_name", "category")

```

```

    .agg(
        sum("sales_amount").alias("daily_revenue"),
        sum("quantity").alias("units_sold"),
        avg("price").alias("avg_price"),
        avg("discount_rate").alias("avg_discount")
    )
)

```

Final Output

Jobs & Pipelines > **cap2_shiv_pipeline** ☆ Send feedback

Development Production Settings Schedule Share Start

Run: 26/8/2025, 2:30:00 PM · Completed Select tables for refresh

Graph List

Pipeline details Update details

Pipeline ID: a31c4a94-c0de-40ff-9852-78e67ad0ca2c

Pipeline type: ETL pipeline

Source code: /Users/tredence19@michaelkloudkrafthotmail.onmicrosoft.com/Cap2_Schema

Run as: Tredence LabsKraft19

Tags: None

Event log Query history

All Info Warning Error Filter...

- 1 minute ago user_action User tredence19@michaelkloudkrafthotmail.onmicrosoft.com started an update.
- 1 minute ago create_update Update 0aecf0 started by USER_ACTION.
- 1 minute ago update_progress Update 0aecf0 is INITIALIZING.
- 1 minute ago unsupported_operation Magic commands (e.g. %py, Show new events) ported with the exception of %pip within a Python notebook. Cells containing magic co..

Pipeline event log details

This event is associated with `cap2_shiv_catalog.cap2_shiv_schema.sales_raw` (Streaming table).

Summary JSON

WARN 2025-08-26 14:36:18 IST flow_progress

Flow 'cap2_shiv_catalog.cap2_shiv_schema.sales_raw' has encountered a schema change during execution and terminated. A new update using the new schema will be automatically started.

Error details

`com.databricks.pipelines.common.errors.DLTSparkException: [FLOW_SCHEMA_CHANGED] Flow cap2_shiv_catalog.cap2_shiv_schema.sales_raw has terminated since it encountered a schema change during execution. The schema change is compatible with existing target schema and the next run of the flow can resume with the new schema.`

`org.apache.spark.sql.streaming.StreamingQueryException: [STREAM_FAILED] Query [id = 6bb00a5f-c7e2-496d-8e48-3230dd79bd92, runId = 3d5bdb32-b6ad-4b42-8d6c-952767db5a28] terminated with exception: [UNKNOWN_FIELD_EXCEPTION.NEW_FIELDS_IN_FILE] Encountered unknown fields during parsing: [Demo], which can be fixed by an automatic retry: true`

Unknown fields inside file path: `abfss://raw@shivghaie.dfs.core.windows.net/sales/sales_day_6.csv`

Rescued file schema: Demo STRING SQLSTATE: KD003 SQLSTATE: XXKST

=== Streaming Query ===

Identifier: `cap2_shiv_catalog.cap2_shiv_schema.__materialization_mat_a31c4a94_c0de_40ff_9852_78e67ad0ca2c_sales_raw_1` [id = 6bb00a5f-c7e2-496d-8e48-3230dd79bd92, runId = 3d5bdb32-b6ad-4b42-8d6c-952767db5a28]

Current Start Offsets: {CloudFilesSource[abfss://raw@shivghaie.dfs.core.windows.net/sales/]: {"seqNum": 7, "sourceVersion": 3, "lastBackfillStartTimeMs": 1756198538801, "lastBackfillFinishTimeMs": 1756198540875, "lastInputPath": "abfss://raw@shivghaie.dfs.core.windows.net/sales/"}}

Current End Offsets: {CloudFilesSource[abfss://raw@shivghaie.dfs.core.windows.net/sales/]: {"seqNum": 9, "sourceVersion": 3, "lastBackfillStartTimeMs": 1756198538801, "lastBackfillFinishTimeMs": 1756198540875, "lastInputPath": "abfss://raw@shivghaie.dfs.core.windows.net/sales/"}}

Current State: ACTIVE

Just now (14s)

9

SQL

```
%sql
SELECT timestamp, level, message
FROM cap2_shiv_catalog.cap2_shiv_schema.event_log
WHERE level = 'ERROR'
ORDER BY timestamp DESC
LIMIT 20
```

(3) Spark Jobs

_sqlcmd: pyspark.sql.connect.dataframe.DataFrame = [timestamp: timestamp, level: string ... 1 more field]

Table +

	timestamp	level	message
1	2025-08-26T09:22:42.079+00:00	ERROR	Update 6c5134 has failed while trying to update table 'cap2_shiv_catalog.cap2_shiv_schema.sales_raw'.
2	2025-08-26T09:20:12.356+00:00	ERROR	Update 8d35e5 has failed while trying to update table 'cap2_shiv_catalog.cap2_shiv_schema.sales_raw'.
3	2025-08-26T09:18:26.966+00:00	ERROR	Update 76a811 has failed while trying to update table 'cap2_shiv_catalog.cap2_shiv_schema.sales_raw'.
4	2025-08-26T08:00:31.687+00:00	ERROR	Update 497569 is FAILED.
5	2025-08-26T07:34:15.380+00:00	ERROR	Update 0da085 is FAILED.

Corrupted File

sales_day_1

sale_id	product_id	region	store_id	quantity	sales_amount	sale_date	Demo
1	157	APAC	Store_31	16	1289.03	2025-08-11 21:20:34	Fvfv
2	65	APAC	Store_9	8	1301.48	2025-08-12 02:40:42	Dfvdvdsav
3	416	LATAM	Store_25	19	3047.56	2025-08-10 00:09:00	Fvdsfav
4	501	US	Store_29	18	1955.74	2025-07-28 09:49:56	Via
5	244	MEA	Store_6	20	2362.98	2025-08-15 00:28:03	Vsdcvads
6	240	EU	Store_22	8	1516.96	2025-08-25 05:58:18	
7	854	EU	Store_24	10	178.92	2025-07-28 09:55:39	
8	69	US	Store_9	7	246.4	2025-08-10 22:41:18	
9	654	EU	Store_3	14	138.23	2025-08-02 05:50:22	
10	277	US	Store_8	15	145.13	2025-08-15 19:05:21	22
11	584	LATAM	Store_4	13	2516.03	2025-08-08 13:20:55	
12	432	APAC	Store_1	13	241.39	2025-07-27 22:24:44	-11
13	797	US	Store_25	11	1914.19	2025-08-05 00:57:54	
14	64	MEA	Store_32	15	2545.81	2025-08-09 06:41:32	Wfwdcvfsad
15	256	US	Store_15	18	1482.75	2025-07-31 09:45:00	
16	924	LATAM	Store_14	7	670.92	2025-08-17 22:33:58	646
17	245	US	Store_38	10	246.84	2025-07-31 15:43:06	
18	779	LATAM	Store_23	6	174.22	2025-07-29 16:52:39	
19	792	MEA	Store_13	7	435.92	2025-08-05 01:06:39	
20	404	APAC	Store_8	10	1822.53	2025-08-05 03:28:00	

Pipeline output with Corrupted File

cap2_shiv_pipeline

DevelopmentProductionSettingsScheduleShareStart

Run: 26/8/2025, 2:35:12 PM Canceled This update has already finished Show the latest update

GraphList

Streaming table

pro_csv

Completed - 6s

View

products

Materialized view

products_clean

Completed - 6s

Streaming table

pro_json

Completed - 6s

Streaming table

sales_raw

Skipped

Materialized view

sales_clean

Skipped

Materialized view

sales_summary

Skipped

Pipeline details

Update details

Pipeline ID

a31c4a94-c0de-40ff-9852-78e67ad0ca2c

Pipeline type

ETL pipeline

Source code

/Users/tredence19@michaelkloudkrafthotmail.onmicrosoft.com/Cap2_Schema

Run as

Tredence LabsKraft19

Tags

None

Event log

Query history

AllInfoWarningError

Filter...

1 minute ago

user_action

User tredence19@michaelkloudkrafthotmail.onmicrosoft.com started an update.

1 minute ago

create_update

Update 6fbb8f started by USER_ACTION.

1 minute ago

update_progress

Update 6fbb8f is INITIALIZING.

1 minute ago

unsupported_operation

Magic commands (e.g. %sql) are not supported with the exception of %%sql within a Python notebook. Cells containing magic co

Show new events

Automate Error Handling

(i) Bronze Layer

```
import dlt
from pyspark.sql.functions import *
# Ingest raw sales data (multiple daily CSV files in raw/sales/)
@dlt.table(
    name="sales_raw",
    comment="Raw sales data ingested from CSV files",
    table_properties={"quality": "bronze"}
)
def sales_raw():
    return (
        spark.readStream.format("cloudFiles")
        .option("cloudFiles.format", "csv")
        .option("header", "true")
        .option("cloudFiles.inferColumnTypes", "true")
        .option("cloudFiles.schemaEvolutionMode", "addNewColumns")
        .option("cloudFiles.schemaLocation", "abfss://raw@shivghaie.dfs.core.windows.net/_schemas/sales/")
        .load("abfss://raw@shivghaie.dfs.core.windows.net/sales/")
        # Force sale_id to string regardless of source type
        .withColumn("sale_id", col("sale_id").cast("string"))
    )

# Products CSV
@dlt.table(
    name="pro_csv",
    comment="Raw product data from CSV",
    table_properties={"quality": "bronze"}
)
def pro_csv():
    return (
        spark.readStream.format("cloudFiles")
        .option("cloudFiles.format", "csv")
        .option("header", "true")
        .load("abfss://raw@shivghaie.dfs.core.windows.net/products/")
    )

# Products JSON
@dlt.table(
    name="pro_json",
    comment="Raw product data from JSON",
    table_properties={"quality": "bronze"}
)
def pro_json():
    return (
        spark.readStream.format("cloudFiles")
        .option("cloudFiles.format", "json")
```

```

        .load("abfss://raw@shivghaie.dfs.core.windows.net/products/")
    )
# Unified Products (CSV + JSON)
@dlt.view(name="products")
def products():
    return dlt.read("pro_csv").unionByName(dlt.read("pro_json"), allowMissingColumns=True)

```

(ii) Silver layer

```

# Clean Sales
@dlt.table(
    name="sales_clean",
    comment="Cleaned sales data with enforced schema",
    table_properties={"quality": "silver"}
)
@dlt.expect("valid_sales_amount", "sales_amount >= 0")
@dlt.expect("valid_quantity", "quantity > 0")
def sales_clean():
    return (
        dlt.read("sales_raw")
        .withColumn("sale_id", col("sale_id").cast("string"))
        .withColumn("product_id", col("product_id").cast("string"))
        .withColumn("region", col("region").cast("string"))
        .withColumn("store_id", col("store_id").cast("string"))
        .withColumn("quantity", col("quantity").cast("int"))
        .withColumn("sales_amount", col("sales_amount").cast("double"))
        .withColumn("sale_date", to_date(col("sale_date"), "yyyy-MM-dd"))
        .na.drop()
    )
@dlt.table(
    name="products_clean",
    comment="Cleaned products data with schema evolution support",
    table_properties={"quality": "silver"}
)
def products_clean():
    return (
        dlt.read("products")
        .withColumn("product_id", col("product_id").cast("string"))
        .withColumn("product_name", col("product_name").cast("string"))
        .withColumn("category", col("category").cast("string"))
        .withColumn("price", col("price").cast("double"))
        .withColumn("discount_rate", col("discount_rate").cast("double"))
    )

```

(iii) Gold Layer

```

# Join sales + products and calculate KPIs
@dlt.table(
    name="sales_summary",

```

```

comment="Daily sales summary by product and region",
table_properties={"quality": "gold"})
def sales_summary():
    sales = dlt.read("sales_clean")
    products = dlt.read("products_clean")
    return (
        sales.join(products, "product_id", "left")
        .groupBy("region", "sale_date", "product_id", "product_name", "category")
        .agg(
            sum("sales_amount").alias("daily_revenue"),
            sum("quantity").alias("units_sold"),
            avg("price").alias("avg_price"),
            avg("discount_rate").alias("avg_discount")
        )
    )
)

```

Corrected Output

Jobs & Pipelines >

cap2_shiv_pipeline ☆ Send feedback

Run: 26/8/2025, 2:56:17 PM · Completed Select tables for refresh

Graph List

Pipeline details Update details

Pipeline ID: a31c4a94-c0de-40ff-9852-78e67ad0ca2c

Pipeline type: ETL pipeline

Source code: /Users/tredence19@michaelcloudkrafttho
tmail.onmicrosoft.com/Cap2_Schema

Run as: Tredence LabsKraft19

Tags: None

Event log Query history

All Info Warning Error Filter...

42 minutes ago	flow_progress	Flow 'cap2_shiv_catalog.cap2_shiv_schema.sales_summary' is STARTING.
42 minutes ago	flow_progress	Flow 'cap2_shiv_catalog.cap2_shiv_schema.sales_summary' is RUNNING.
41 minutes ago	flow_progress	Flow 'cap2_shiv_catalog.cap2_shiv_schema.sales_summary' has COMPLETED.
41 minutes ago	update_progress	Update cb20ae is COMPLETED.

Pipeline event log details

This event is associated with `cap2_shiv_catalog.cap2_shiv_schema.sales_summary` (Materialized view).

Summary JSON

INFO 2025-08-26 14:58:52 IST flow_progress

Flow 'cap2_shiv_catalog.cap2_shiv_schema.sales_summary' has COMPLETED.

Just now (5s)

9

SQL

```
%sql
-- Check if new rows landed in silver (clean data)
SELECT *
FROM cap2_shiv_catalog.cap2_shiv_schema.sales_clean
WHERE sale_date = '2025-08-26';
--Check Gold aggregation
SELECT *
FROM cap2_shiv_catalog.cap2_shiv_schema.sales_summary
WHERE sale_date = '2025-08-26';
--Check event log again (should now show success)
SELECT timestamp, level, message
FROM cap2_shiv_catalog.cap2_shiv_schema.event_log
ORDER BY timestamp DESC
LIMIT 20;
```

▶ (1) Spark Jobs

▶ _sqldf: pyspark.sql.connect.dataframe.DataFrame = [timestamp: timestamp, level: string ... 1 more field]

Table +

Q 🔍 ⌵ □

	timestamp	level	message
1	2025-08-26T09:28:54.383+00:00	INFO	Update cb20ae is COMPLETED.
2	2025-08-26T09:28:54.382+00:00	METRICS	Reported flow time metrics for flowName: 'pipelines.flowTimeMetrics.missingFlowName'.
3	2025-08-26T09:28:54.380+00:00	METRICS	Reported flow time metrics for flowName: 'cap2_shiv_catalog.cap2_shiv_schema.products_clean'.
4	2025-08-26T09:28:54.378+00:00	METRICS	Reported flow time metrics for flowName: 'cap2_shiv_catalog.cap2_shiv_schema.sales_summary'.
5	2025-08-26T09:28:52.454+00:00	INFO	Flow 'cap2_shiv_catalog.cap2_shiv_schema.sales_summary' has COMPLETED.
6	2025-08-26T09:28:50.194+00:00	METRICS	Reported cluster resources metrics.
7	2025-08-26T09:28:35.763+00:00	METRICS	> 'cap2_shiv_catalog'. 'cap2_shiv_schema'. '__materialization_mat_a31c4a94_c0de_40ff_9852_78e67ad0ca2c'
8	2025-08-26T09:28:32.973+00:00	INFO	Flow 'cap2_shiv_catalog.cap2_shiv_schema.sales_summary' is RUNNING.
9	2025-08-26T09:28:32.941+00:00	INFO	Flow 'cap2_shiv_catalog.cap2_shiv_schema.sales_summary' is STARTING.
10	2025-08-26T09:28:32.937+00:00	INFO	> Flow 'cap2_shiv_catalog.cap2_shiv_schema.sales_summary' has been planned in DLT to be executed as COMPL

Time travel

%sql

```
SHOW TABLES IN cap2_shiv_catalog.cap2_shiv_schema;
```

```
CREATE OR REPLACE TABLE
```

```
cap2_shiv_catalog.cap2_shiv_schema.sales_clean_tbl AS
```

```
SELECT * FROM cap2_shiv_catalog.cap2_shiv_schema.sales_clean;
```

```
DESCRIBE HISTORY cap2_shiv_catalog.cap2_shiv_schema.sales_clean_tbl;
```

```
-- Query older version
```

```
SELECT *
```

```
FROM cap2_shiv_catalog.cap2_shiv_schema.sales_clean_tbl VERSION AS OF 0
```

```
LIMIT 10;
```

The screenshot shows a Databricks SQL interface. At the top, a status bar indicates 'Just now (15s)' and '10' rows. The SQL query is displayed in a dark-themed editor. Below the query, a message states '(3) Spark Jobs'. A table header is visible with columns: sale_id, product_id, region, store_id, quantity, sales_amount, sale_date, and Demo. The table body is empty, displaying 'No rows returned'. At the bottom, a status bar shows '0 rows | 14.81s runtime' and 'Refreshed now'. A footer note states: 'This result is stored as _sqldf and can be used in other Python and SQL cells.'

```
%sql
SHOW TABLES IN cap2_shiv_catalog.cap2_shiv_schema;

CREATE OR REPLACE TABLE
cap2_shiv_catalog.cap2_shiv_schema.sales_clean_tbl AS
SELECT * FROM cap2_shiv_catalog.cap2_shiv_schema.sales_clean;

DESCRIBE HISTORY cap2_shiv_catalog.cap2_shiv_schema.sales_clean_tbl;
-- Query older version
SELECT *
FROM cap2_shiv_catalog.cap2_shiv_schema.sales_clean_tbl VERSION AS OF 0
LIMIT 10;
```

Table

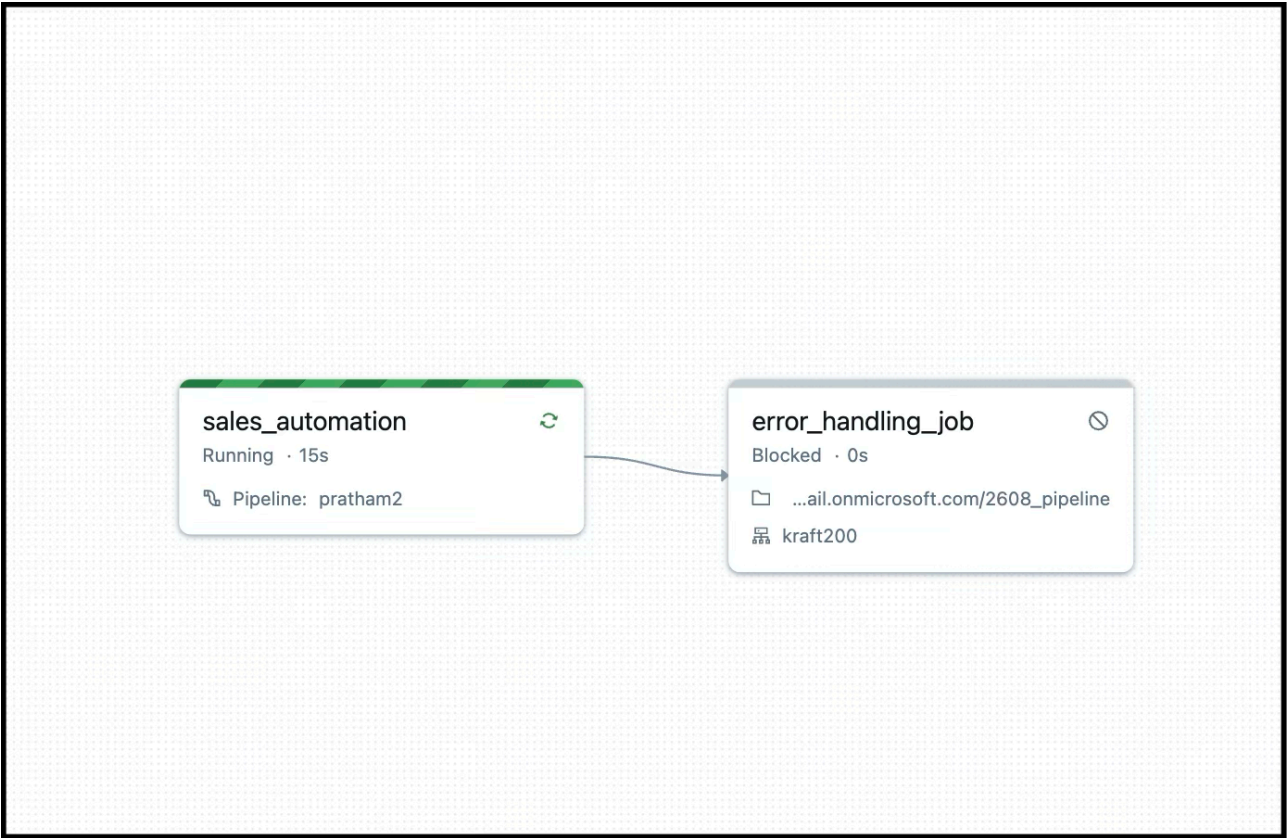
sale_id	product_id	region	store_id	quantity	sales_amount	sale_date	Demo
No rows returned							

0 rows | 14.81s runtime

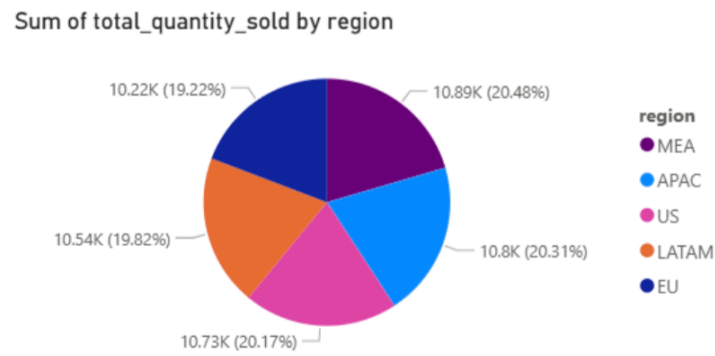
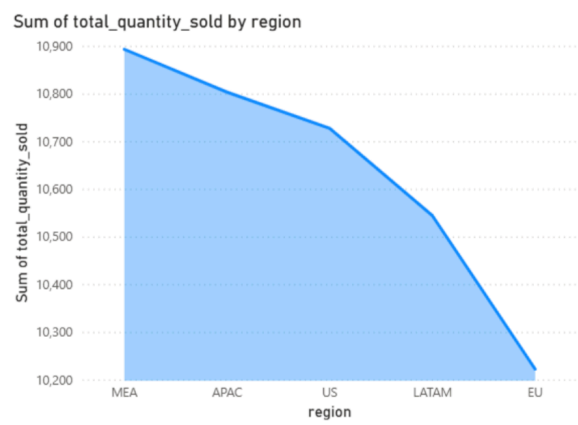
Refreshed now

This result is stored as `_sqldf` and can be used in other Python and SQL cells.

Creation of Automation Workflow Job



Power BI



Sum of total_quantity_sold by region

● Increase ● Decrease ● Total



Sum of total_quantity_sold by product_id

